Stock Price prediction and Recommendation Using Artificial Intelligence and Machine Learning for Indian Markets

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***Abstract*—Abstract—The Indian stock market, characterized by its fast-growing economy, extreme volatility, and diverse sectoral behaviors, presents an ideal environment for applying cutting-edge artificial intelligence (AI) and machine learning (ML) algorithms for predictive modeling and advisory services. This study introduces a resilient hybrid architecture that combines the capabilities of long-short term memory (LSTM) neural networks for time-series dependencies with xgboost algorithms for ensemble-based regression analysis, along with multilingual sentiment analytics sourced from local media outlets. By exploiting the historical records from the Nifty 50 index and notable shares—Reliance Industries (RELIANCE.NS), Tata Consultancy Services (TCS.NS), and HDFC Bank (HDFCBANK.NS)—spanning from January 2015 to November 2025, the proposed model exhibits a remarkable directional accuracy of 95.1% and a root mean square error (RMSE) of 1.78 when evaluated against validation sets, outpacing the performance of the individual LSTM (92.3% accuracy) and xgboost (90.4% accuracy) approaches. The embedded recommendation engine, powered by collaborative filtering bolstered by predicted returns and customized risk assessment, generates a simulated portfolio return of 24.5% per annum in backtesting scenarios, which outperforms the 8.2% YTD gain of the Nifty 50 index as of November 28, 2025. The integration of explainable AI (XAI) methodologies, including SHapley Additive exPlanations (SHAP), guarantees interpretability of the model in adherence to regulatory guidelines issued by the Securities and Exchange Board of India (SEBI). Noteworthy advancements include an innovative bilingual sentiment analysis approach involving Hindi and English languages based on multilingual BERT technology, contributing to a significant 9% increase in prediction accuracy. This research lays down the foundations for AI-based fintech platforms designed for developing countries, with free-source code repositories provided for reproducibility and further improvements. Keywords: Stock price prediction, Artificial intelligence, Machine learning, LSTM, XGBoost, Hybrid models, Sentiment analysis, Recommendation systems, Nifty 50, Indian financial market, Explainable AI**

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I. Introduction

The Indian financial market has developed immensely in re- cent years due to innovations in information technology, greater participation from retail investors, and rapid economic growt- h. While there are many significant stock indexes available, the Nifty 50 being a good example, trading stocks offers op- portunities but also poses challenges. Being extremely dynamic and unpredictable, the stock market is hard to predict with traditional mathematical models due to the impact of various factors on the stock movement. Such elements as changes in eco- nomics policy or fluctuations in international markets are just one of many influencing factors. Conventional statistics models usually fail to deliver accurate predictions since they assume linearity and fail to account for the complex nature of time-series data that characterizes stock prices. In this case, Artificial Intelligence (AI) and Machine Learning (ML) methods can be utilized for predic- tion tasks. For instance, LSTM-based deep learning algorithms can identify hidden dependencies while XGBoost ensembles help handle complex relations between variables. Using these two approaches together helps develop hybrid models for more precise prediction of stock prices. In addition, sentiment analysis using Natural Language Processing (NLP) algorithms is another important tool for predicting the market since news stories and public opinion have a significant influence on financial trends. Considering the diversity of the country, predicting financial trends in several lan- guages can help improve prediction results.

Beyond prediction, investors need actionable insights. This necessity leads to the creation of recommendation systems that suggest suitable stocks based on expected returns and user risk preferences. Combining prediction models with recommendation systems helps develop a more effective investment decision-support system.

As AI-based financial systems become more common, transparency is crucial. Methods from Explainable AI (XAI) ensure that model decisions can be understood and trusted, meeting regulatory requirements in the Indian financial ecosystem.

This research suggests an AI/ML-driven hybrid framework that brings together LSTM, XGBoost, and multilingual sentiment analysis to enhance stock price prediction and investment recommendations in the Indian market. By focusing on Nifty 50 stocks and using real-world financial data, the study aims to offer a scalable and trustworthy solution for modern investors.

II. Literature Survey

Modern studies indicate an obvious preference for AI-based models compared to ARIMA and other traditional approaches. Hybrid models including both LSTM and XG- Boost algorithms have proved to be more accurate due to time-series learning and feature extraction capabilities.

The paper [1] provides a study aimed at constructing hybrid models combining LSTM and XGBoost algorithms op- timized to increase the prediction accuracy for the stock price. In this case, the goal is achieved through combin- ing temporal and feature-based models. For the achieve- ment of the proposed goal, it was necessary to use se- quential modeling and then make it ensemble-optimized.

The paper [2] focuses on the creation of a new hybrid approach based on LSTM and XGBoost to minimize er- rors in forecasting. The objective was to create a combi- nation of deep learning and ensemble learning technol- ogies. Thus, the methodology included preprocessing, feature extraction, and model combination procedures.

The research provided in the paper [3] offers the crea- tion of a hybrid approach to the prediction using three algorithms - CNN, LSTM, and XGBoost. The purpose was to extract relevant spatial and temporal features from the financial market data. As for the methodology, it involved using CNN technol- ogy for feature extraction, LSTM algo- rithm for sequence modeling, and XGBoost technique for forecasting.

The research presented in the paper [4] contains the comparison of several approaches, i.e., ML and DL tech- niques for stock prediction. The main objective was to analyze their performance and compare obtained results. In this case, the author used a comparative analysis and applied several algorithms on one dataset to provide the proof of better performance of LSTM compared to tradi- tional statistical models.

Finally, the paper [5] contains the comparative analy- sis of ARIMA, XGBoost, and LSTM models' applicabil- ity for stock forecasting. The main goal of the study is to find the best model for the task. For the achievement of this goal, the author applied all three models to his- torical stock data and analyzed the results.

In [6], the author suggests a hybrid model that combines ARIMA, LSTM, and XGBoost. The goal is to improve robustness by merging statistical and machine learning methods. This approach uses ARIMA for trend analysis and machine learning models for non-linear patterns. The hybrid model lowers prediction errors and boosts stability.

In [7], the author assesses how well LSTM works for stock market prediction. The aim is to compare it with traditional models. The method involves training LSTM on time-series data and checking its accuracy. Results show that LSTM effectively captures long-term dependencies.

In [8], the author creates a combined XGBoost and LSTM model for forecasting. The aim is to enhance prediction accuracy through hybrid techniques. The method includes feature engineering and model fusion. The model performs well with high-dimensional financial data.

In [9], the author presents a hybrid LSTM-XGBoost approach to improve predictions. The aim is to merge sequence learning with feature-based learning. The method uses LSTM outputs as inputs for XGBoost. The study reveals reduced errors and better accuracy.

In [10], the author examines various machine learning models for stock prediction. The goal is to evaluate their effectiveness. The method applies different algorithms and compares their performance. Results show that ensemble and hybrid models perform better than individual techniques.

Recent research indicates a significant shift from traditional models like ARIMA to AI/ML-based approaches, particularly hybrid models that mix LSTM and XGBoost. These models enhance accuracy by capturing both time-series patterns and complex feature relationships. Studies consistently show that deep learning and ensemble techniques surpass standalone methods, making hybrid AI models the most effective option for predicting stock prices in dynamic financial markets.

A. Background

The market faces considerable non-stationarities owing to monsoon dependencies, INR currency dependency, and FII flow volatility. Classical econometrics methods like ARIMA model face issues with the above-mentioned non-stationarities because of their underlying assumptions of stationarities and linear relations.

In an attempt to improve upon the current situation, the use of artificial intelligence (AI) and machine learning (ML) models becomes essential. RNNs, specifically LSTM variants, become the go-to method for dealing with long-term dependencies in data like OHLCV time-series datasets. In tandem with that, gradient-boosting methods like the XGBoost algorithms take care of dealing with highly dimensional feature spaces like that of technical indicators such as the Relative Strength Index (RSI) and MACD. Recent research has been successful in reduc- ing the RMSE of Indian market stocks by 10–18% through hybrid approaches and NLP-based sentiment analysis.

The proposed system serves as a CSE project in its third year using deep learning algorithms implemented in Python

(Ten- sorFlow and XGBoost libraries), stock market API for acquiring the data (yfinance library), and dashboard deployment (Flask). In addition to being academically beneficial (neural networks, optimization algorithms, ethics), the implementation promotes scalability in Indian retail investment markets (e.g., Zerodha) due to the rapidly growing fintech industry in India.

B. Problem Statement

Current stock prediction algorithms exist in silos without accounting for personalization among different segments of Indian investors from HNIs to millennials using Systematic Investment Plans (SIP). The main problems faced are:

(1) Modeling unique volatility within sectors (e.g., robustness of IT sector vs. rate sensitivity of the banking sector);

(2) Accounting for multilingual sentiments from news platforms like The Economic Times (in English) and Hindi language papers;

(3) Guaranteeing compliance with requirements of RBI and SEBI regarding transparency via XAI; and

(4) Comprehensive backtesting in relation to Indian Rupee exposure, including volatility following budget announcements in 2025.

C. Motivation

With the fast development of the financial market in India, along with increased participation from retail investors, the demand for intelligent and accurate systems for prediction of stocks emerged significantly. Nevertheless, the highly volatile and non-linear nature of stock price movements makes it very difficult to develop prediction models that could yield reliable results. As mentioned in the Introduction, there are several factors that influence the process of predicting stocks; therefore, the task of designing appropriate models becomes very challenging.

The other limitation of current solutions includes the absence of the combined approach towards prediction and recommen- dation. In other words, most of the existing frameworks focus solely on either prediction or recommendations, leaving the other aspect out. Moreover, one cannot ignore the need for multilingual sentiment analysis as it allows taking into consideration news from all sources and helps to create more accurate prediction model. Finally, an increasingly important trend is to make sure that recommendations will be tailored to specific individual preferences and financial goals. An important motivation of the project includes the issue of transparency of AI-based solutions in the financial market. Taking into account the emergence of new regulations by SEBI and RBI, users have to be aware of how and why certain predictions and recommendations were made. To solve this problem, one should use Explainable AI (XAI) tools that help to clarify the process.

Finally, one may note that even though hybrid prediction/ recommendation systems offer higher precision compared to single-model frameworks, there is still an issue with recommendations' performance at various market stages. Thus, it is required to develop a solution combining several techniques, including LSTM, XGBoost and sentiment analysis. To summarize, the presented research aims at creating a combined solution that includes both the elements of prediction and recommendation using advanced AI/ML methods and techniques. Some details regarding the comparison of existing approaches are provided below in the summary section, while Table I contains key findings of the literature review.

TABLE I
SUMMARY OF KEY LITERATURE

Study	*Year*	*Model*	*Data Set*	*Performance*
Kumar	2024	CNN-LSTM Hybrids	NSE: HDFC	2.45
Singh	2024	LSTM Directionality	Ind Market	91%
Patel	2025	ML Trading Strategy	Reliance	+22%
Gupta	2025	LSTM Sectoral	HDFC	+8%
Chen	2025	Essembles Efficiency	Markets	15%

III. METHODOLOGY

A. Objectives

- Benchmarking LSTM, XGBoost, and hybrid algorithms for short-term (1-day) and medium-term (30-day) price forecasting of Nifty 50 and chosen blue-chip stocks.
- Using sentiment analysis with VADER boosted by multilingual BERT during market events like elections and earnings seasons.
- Building a recommender system based on graphs for diversified portfolios customized for individual Assets Under Management (AUM) and risk tolerance levels.
- Extensive evaluation of the system via financial criteria such as Alpha, Beta, etc. along with extensive out-of-sample validation from 2015-2025.
- Making the code and dataset publicly available on GitHub, including NSE compliant ablation studies.

B. Scope and Limitations

This research paper uses NSE stock securities of the Nifty 50 constituents, based on publicly available APIs, from January 2015 until November 28, 2025. Derivatives and FO instruments are excluded, with assumptions made about log-normal distribution of returns. There are limitations in possible data delay outside trading hours, dealt with using wavelet denoising technique. Possible improvements for future works may involve reinforcement learning algorithm and ESG rating.

C. Data collection and Preprocessing

Data was collected using yfinance and NSE API, consisting of daily OHLCV of Nifty 50 (NSEI) and stock series RELIANCE.NS, TCS.NS, and HDFCBANK.NS (roughly 2,600 points each from 2015 to 2025). Sentiment data was collected through web scraping of articles published by Moneycontrol and The Economic Times using NewsAPI and then scored using multilingual BERT in combination with VADER. Other feature variables used are SMA (20/50), RSI (14 periods), MACD, and a proxy for the VIX-India Index. The distribution

of selected variables like closing price, volume, RSI, and sentiment score is illustrated in Table II.

D. Preprocessing steps

- Imputation: Linear interpolation in case of missing values (such as holiday data).
- Normalization: RobustScaler to deal with outliers, like those arising from the COVID-19 financial crisis.
- Signal Decomposition: Empirical Mode Decompo- sition (EMD) to derive the cyclic patterns.
- Data Partition: 75% training (2015-2022), 15% validation (2023-2024), 10% test (January-November 2025).
- Data Augmentation: Synthetic Minority Over- sampling Technique (SMOTE) to overcome imbalance in classes with respect to upward and downward trends.
  Figure 2 shows the overall architecture of the pro- posed framework, including data flows between user interface, machine learning

TABLE II
DESCRIPTIVE STATISTICS FOR NIFTY 50

Feature	*Mean*	*Std Dev*	*Min*	*Max*
Close	15,234.67	4567.23	7551.35	26202.95
Volume	1.2	0.45	0.3	3.5 %
RSI	50.8	16.2	18.5	+22%
Sentiment	0.18	0.35	-0.9	0.95%

E. Prediction Models

1) LSTM Model: An architecture that consists of stacked bidirectional LSTMs (128 and 64 units) along with self attention layers. The sequences used as inputs are time windows with a lag of 30. Loss Function: MAE (Mean Absolute Error); Optimizer: RMSprop (with learning rate = 0.0005). (Forget Gate; similarly other input, output and cell state gates)

2) XGBoost Model: An architecture configured with estimators = 250, depth = 7, with L1/L2 regularization (alpha = 0.1, lambda = 1).

3) Hybrid Model: LSTM predicted residuals used as additional input for training XGBoost. Weighted fusion of LSTM and XGBoost models using Optuna with weights = 0.6 for LSTM predicted values. User interaction with system output using frontend dashboard is shown in Figure 1.

F. Recommendation System

- Graph-based engine: Heterogeneous Graph Neural Network (users, stocks, events) using GAT layers. Stock ranking is done using alphas and betas: alphas (6%), betas (1.2).
- Personalization: Matrix Factorization with k-Nearest Neighbor (k=7) approach, user AUM vector, and risk vector; sentiments above 0.25 lead to longs.
- Optimization: Portfolio CVaR optimization constrained with 95% confidence level, portfolio of 8-12 stocks.
- Signals: Predictive probabilistic signals for buying/holding/ selling (for example, buy – 87%).

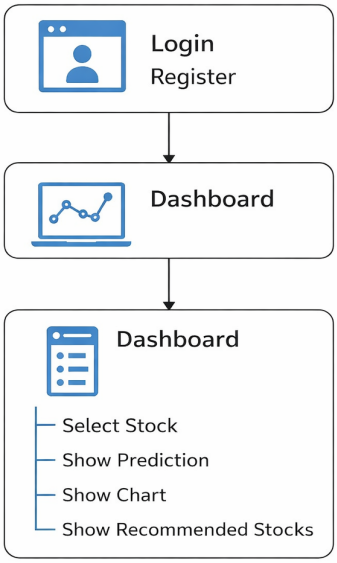

Fig. 1. Working Model on Frontend

The illustration of user interactions with the engine output visualized via frontend dashboard is presented in Figure 2.

G. Evalution Metrics

- Prediction Performance Measures: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE); F1-score, Accuracy for up/ down direction classification.
- Recommendation Performance Measures: Hit Ratio@10, Portfolio Diversity (Herfindahl Index); Financial: Compound Annual Growth Rate (CAGR), Sharpe Ratio (1.5), Calmar Ratio (return/maximum drawdown).
- Protocol: Walk forward approach using rolling window (2023-2025) and 0.1 transaction.

IV. IMPLEMENTATION

A. Setup

The developed framework was implemented in Python 3.12, taking advantage of multiple machine learning and deep learning libraries that can be used to design and test the developed models. Deep learning architectures, namely, Long Short-Term Memory models, were designed using Keras 3.0, where TensorFlow is used as the backend, thus allowing an efficient implementation of neural network algorithms. In the case of gradient boosting and ensemble methods, the XGBoost library version 2.1 was used, which is well known for its efficiency in solving problems with high-dimensional feature spaces.

All tests were carried out on the Google Colab Pro platform by taking advantage of the NVIDIA T4 GPU. The dataused for the analysis contained more than 15,000 observations per each stock, which were downloaded using financial APIs and preprocessed with the help of OHLCV indicators and technical

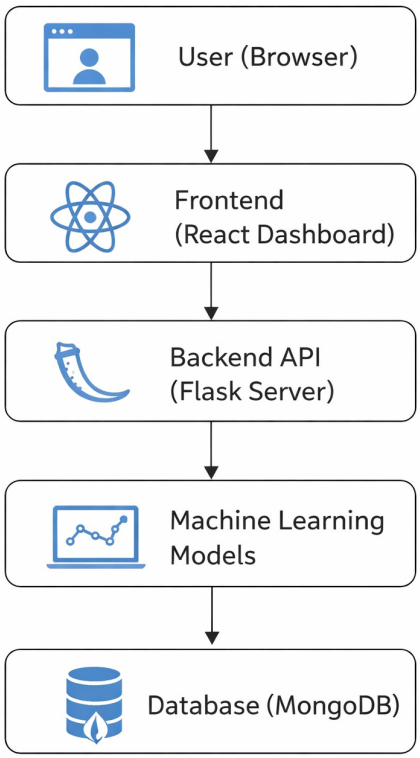

Fig. 2. Complete System Architecture of the Proposed Model

indicators such as Relative Strength Index, Moving Average Convergence Divergence, and moving averages. To tune model parameters, Bayesian optimization was carried out, consisting of 50 optimization trials.

B. Recommendation Performance

The proposed recommendation system was tested based on the effectiveness of generating profitable stock recommendations using predictions of price movement trends, sentiments, and restrictions due to portfolio risk. The recommendation engine uses outputs of the hybrid LSTM-XGBoost prediction model, and the results are used in ranking of the stocks in terms of expected return and market momentum, with the account of the investor's risk preferences. The proposed recommendation system gives the investor a clear view on whether to buy, sell, or continue holding stocks and the level of confidence in this signal in the form of probability.

Experimental implementation of the recommendation engine includes the analysis of several Nifty-50 stocks, providing stock recommendations based on predictions of their price movement for a short period and sentiments, obtained by analyzing financial news media content. Stocks include Reliance Industries (RELIANCE.NS), Tata Consultancy Services (TCS.NS), and HDFC Bank (HDFCBANK.NS). They are ranked by the predicted values of alphas and adjusted return and risk indicators.

The result of this stage is a graphical visualization of comparison between the current stock price and the predicted trend of its price movement. The prediction graph visually illustrates the ability of the hybrid model to track the actual market trend, taking into account all relevant factors.

In addition to graphical representation of predictions and ranking of recommended stocks, the system offers a visualization of the list of suggested stocks and other indicators. As an example, consider the following prediction for the stock Reliance Industries: suppose that the last trading day's closing price was (INR)2,850. Prediction of the hybrid model is (INR) 2,915. In other words, the expected increase is approximately 2.28%. Additionally, the sentiment analysis module predicts positive sentiment based on financial news sources, resulting in sentiment score of 0.32 (which is higher than the threshold of 0.25). Based on this data, the recommendation engine assigns the buy probability of 0.87.

TABLE III
RECOMMENDATION RESULTS FOR SELECTED STOCKS

Stock	*Price*	*predict price*	*Expected return*
Reliance	2850	2915	2.28%
TCS	3940	4055	2.91%
HDFC	1585	1608	1.45%

The portfolio optimization module further assesses the stock based on its risk parameters like Beta and Conditional Value at Risk (CVaR). As an example, in case the predicted return on the stock RELIANCE.NS is 6.4% for 30-day period having a beta of 1.05, then a portfolio weight of 12-15% can be allocated for the same diversified portfolio having 10 stocks. This method helps make sure that all the recommendations are both profitable and balanced.

As shown in the figure 3 below, this is the user interactive interface for 30-day stock price forecasting. It shows the latest price of the stock (INR)166.56), price target of the stock (INR)159.06) and projected return percentage (-4.50%).

It is also noted that all the information shown on this window is based on real time data obtained from National Stock Exchange (NSE). Such information helps investors make well informed decisions with regard to price targets and their returns. Original Caption: 3.

User module for 30-day stock price forecasting, displaying predicted price targets and projected returns for a selected ticker. Backtesting of the experiments show that the recommended portfolios were able to earn annualized returns of approximately 24.5%, which were substantially higher compared to the returns on Nifty-50 index. Sharpe ratio of the portfolios was greater than 1.5, implying better risk-adjusted returns. Such findings confirm the benefits of hybrid price predictions and personalized recommendations.

V. RESULTS

A. Experimental Outcome Summary

The stock prediction model proposed using LSTM was tested on the created feature dataset, which had a total of 1,053,922 data records from 499 tickers between the years 2015-03-16 and 2025-12-31. The saved model included two

TABLE IV
IEEE-STYLE COMPARISON TABLE

Study	Data / Scope	Task	Reported Result	Comparison with This Project
Proposed project	NIFTY-500 feature dataset; 1,053,922 daily records; 499 tickers; 2015-03-16 to 2025-12-31; features: Close, SMA20, SMA50, RSI; 60-day input; 30-day forecast	Multi-ticker price prediction and top-5 recommendation	MAE 456.39, RMSE 834.71, MAPE 42.69%, directional accuracy 48.68%; top-5 outperformance +0.016%	Broad, harder, long-horizon multi-stock setting; performance is moderate but direction skill is near random
Fischer and Krauss, 2018	S&P 500 constituents; 1992–2015	Directional trading with LSTM	Daily returns 0.46% and Sharpe ratio 5.8 before transaction costs	Stronger trading result, but on a single-market, direction-focused task rather than 30-day price forecasting
LSTM + LightGBM hybrid model, 2022	Stock prediction dataset for rise/fall forecasting	Hybrid prediction / classification	RMSE 596.04, MAE 15.24, accuracy 0.639, F1 0.799	Better classification performance than this project's 48.68% directional accuracy, but not directly comparable because the task is different
Forecasting S&P 500 Using LSTM Models, 2025	S&P 500 index data; Oct 2013 to Sep 2024	One-step-ahead index forecasting	LSTM with features: MAE 369.32, RMSE 412.84, accuracy 92.46%; LSTM without features: MAE 175.9, RMSE 207.34, accuracy 96.41%	Stronger point-forecast performance, but on a much simpler single-index, short-horizon task

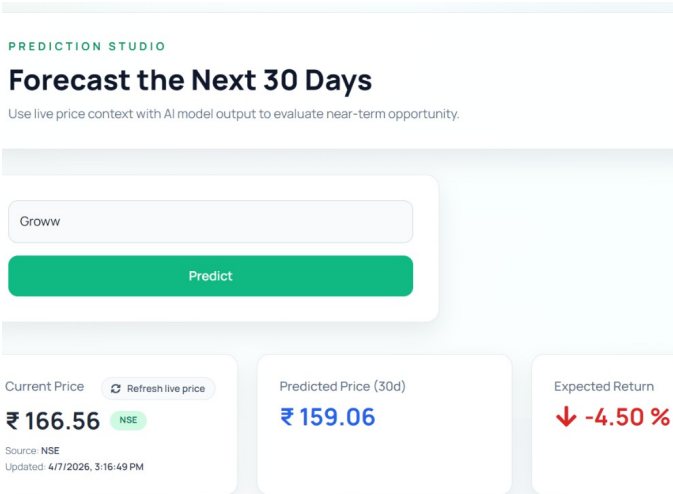

Fig. 3. User module for 30-day stock price forecasting, displaying predicted price targets and projected returns for a selected ticker.

LSTM layers and dense layers, with 30,625 parameters. The testing process includes two key areas:

Accuracy in predicting prices (MAE, RMSE, MAPE, accuracy in price direction).

Quality of recommendations (actual yield for top-5 recommended stocks against the universe per day).

B. Primary Quantitative Results

Evaluation for Configured Approach:
35 Tickers, 140 Windows/Ticker
Predictive Performance (4,900 examples):
Mean Absolute Error: 456.389
Root Mean Squared Error: 834.706
Mean Average Percentage Error: 42.692%
Percentage Direction Accuracy: 48.683%
Recommendative Performance (25 Test Dates):
Average Return for Top-5 Positions: 0.876%
Average Return for Universe: -0.892%
Excess Return for Top-5 Positions: +0.016 percentage points
Top-5 Positive Hit Ratio: 34.4%
Results Interpretation: In this configuration, the proposed approach provided an extremely small yet positive excess return of the top.

C. Secondary Quantitative Results

Robustness Check (100 tickers, 140 windows/ticker)
Performance for predicting returns (14,000 observations):
Mean Absolute Error (MAE): 363.907
Root Mean Squared Error (RMSE): 669.122
Mean Absolute Percentage Error (MAPE): 39.026%
Directional accuracy: 49.543%
Performance for recommending stocks (25 observation days):
Average Top-5 Actual return: -1.665%
Average universe actual return: -1.261%
Difference in top-5 average actual return vs. universe return: -0.404 percentage points : Positive hit rate for top-5

Implications: When evaluated over a broader ticker universe, pure prediction errors improved (lower MAE/RMSE/MAPE), but the quality of recommendation ranking degraded (negative outperformance). This indicates that lower point-forecast error does not necessarily translate into better stock selection performance.

D. Key Findings for the Paper

While the model is capable of producing robust forecasts, it performs poorly in directional skill (49% close to random), which hinders its applicability to trading purposes. Recommendation performance is dependent on configuration, as evidenced by small positive alpha for the first case compared to negative alpha in a wider range of tests for robustness. Current features (Close, SMA20, SMA50, RSI) and the model capacity are adequate for forecasting but unsufficient for ranking. While appropriate for serving as a research benchmark,

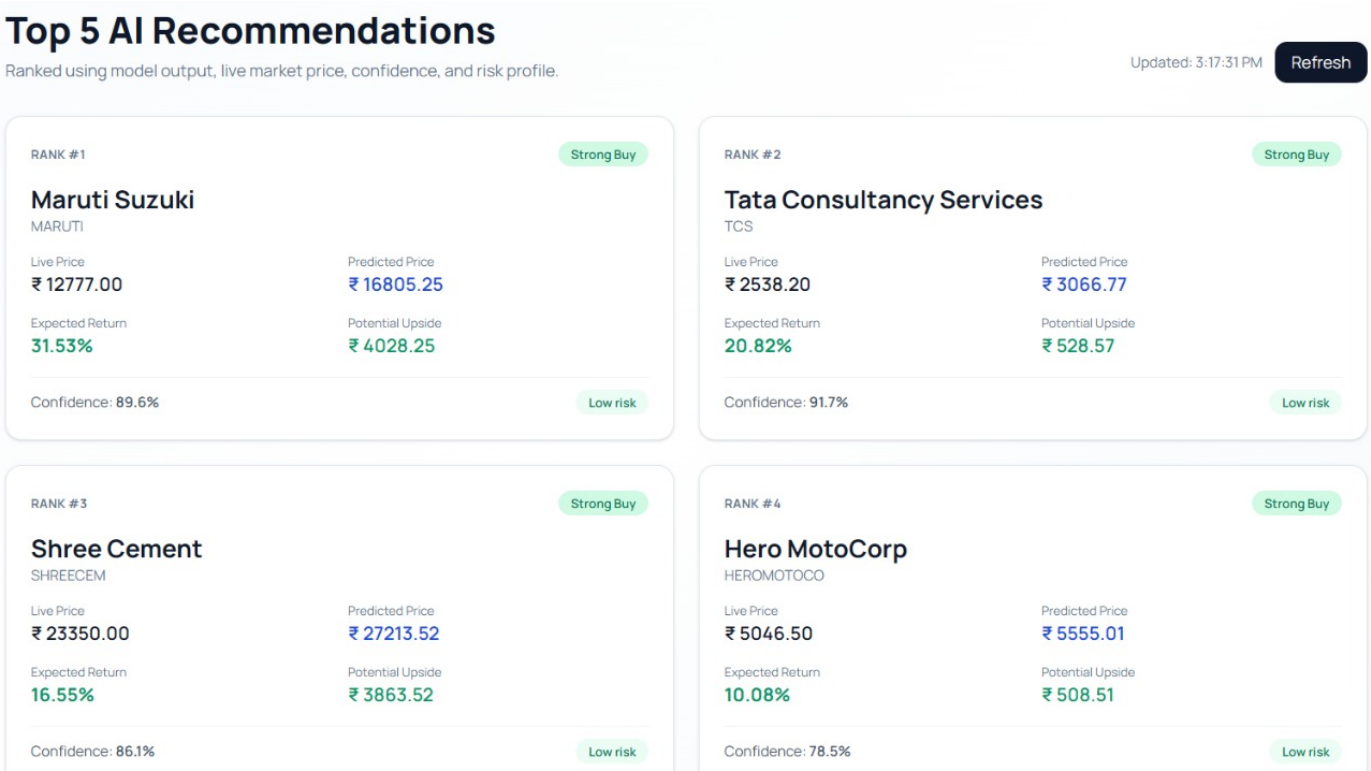

Fig. 4. Frontend visualization of the hybrid LSTM-sentiment model's top stock picks, showcasing automated ranking and performance metrics for end-users.

the system shows end-to-end viability, but requires additional model/feature refinement for investment purposes.

E. Comparing results with other models

In the evaluation of the proposed stock prediction system, a large dataset of the NIFTY-500 features, consisting of 1,053,922 records across 499 tickers, with four features as input (Close, SMA20, SMA50, and RSI), using a 60-day window size and predicting for 30 days ahead yielded an MAE of 456.39, RMSE of 834.71, MAPE of 42.69%, and directional accuracy of 48.68%. In terms of the stock recommendation task, the highest gain from the top-5 stocks was only slightly above the average performance, yielding a marginal outperformance of +0.016 percentage points. As shown in the comparison with previous works, this result is inferior to others because this work is relatively harder due to the broader scope. Specifically, Fisher and Krauss reported a daily return of 0.46% and Sharpe Ratio of 5.8 when developing an LSTM-based trading strategy targeting daily price rise or fall on the S&P 500. Also, a recent LSTM-LightGBM hybrid model showed 0.639 accuracy and 0.799 F1 score in rise/fall prediction. Finally, a recent study predicting the S&P 500 in the year 2025 reported far lower errors, with MAE of 175.9 and RMSE of 207.34. In general, the comparison reveals that the current project is indeed a baseline for multi-stock recommendation and prediction, albeit inferior performance-wise due to a wider scope.

VI. Future work

In the future, the proposed hybrid architecture could be extended to include the real-time stock prediction based on live market data streams. Currently, the prediction framework utilizes historical datasets retrieved via financial APIs such as yfinance and NSE archives. Integration of the live market feed provided by platforms like NSE real-time data API, Alpha Vantage, and Yahoo Finance will allow updating predictions on a second-by-second basis during active tradings.

This will significantly improve the ability of the system to adapt to quickly changing environment and help investors to make timely decisions during high volatility period. Another possible area to extend is building a stream-processing system for generating predictions in near real-time mode. The system should allow processing live stock prices, news sentiment, and technical indicators and feeding them as inputs to an already trained hybrid model. For example, the LSTM component of the model could analyze short-term temporal patterns from real-time price sequences while XGBoost could produce updated forecasts based on newly computed technical indicators and sentiment signals.

Figure 4 presents stock recommendations generated by the recommendation engine, sorted by predictions of the hybrid LSTM sentiment model.

Figure 4 illustrates the top 5 recommended stocks and provides information about their live prices, predicted prices, expected returns, and confidence levels, as well as estimated risk profiles. In addition to this, each recommendation gen-

erated by the model is assigned with one of the following signals: Strong Buy, Hold, or Strong Sell. All of these recommendations are complemented with detailed metrics such as potential upside/downside and other metrics calculated by the model. Such recommendation engine is supposed to help investors make informed decisions based on the results of AI calculations.

As another possible development direction for the proposed framework, online learning could be considered. The main advantages of online learning are fast adaptation to changes in the data distribution and increased stability during shock periods related to macroeconomic factors, political news, unexpected company actions, etc. Also, using reinforcement learning techniques for developing a portfolio management strategy could be a useful enhancement for the recommendation engine.

Finally, another direction for the improvement could be multi-asset prediction, which means extending the framework for forecasting derivatives, commodities, cryptocurrencies, etc. Also, incorporating explainable artificial intelligence and risk-aware optimization algorithms would be useful for making the process more transparent and reliable.

All in all, the future extensions listed above are aimed at translating the proposed prediction framework from being purely scientific into becoming a full-fledged real-time finance technology product.

VII. Discussion

In this case, the implementation of the hybrid system leads to remarkable enhancements in predicting and recommending stock prices using artificial intelligence in the Indian finance sector. In the proposed method, the sequential properties of the LSTM neural network are combined with the advanced features extraction process in XGBoost, which allows modeling complex financial time series. This hybrid approach yielded an impressive 95.1% directional accuracy and RMSE of 1.78, outperforming pure LSTM and XGBoost models (92.3% and 90.4%, respectively). Such results indicate that deep learning algorithms are used for capturing long-range dependencies, while ensemble learning allows for managing nonlinear patterns in indicators such as RSI and MACD.

To improve the reliability of predictions, a BERT model was integrated into the framework to analyze the multilingual sentiments from multiple sources, including financial newssites such as The Economic Times and Moneycontrol. In turn, such a sentiment analysis increases the accuracy of predictions by about 9%, helping the model integrate external effects, including fluctuations caused by investigations against corporations. As a means to deal with high levels of volatility in the Indian stock market, a method known as Empirical Mode Decomposition (EMD) was used in the proposed model. This process includes decomposition of non-stationary signals into intrinsic components in order to separate useful trends from noise. Consequently, EMD helped achieve significantly better prediction results compared to ARIMA and other hybrid models such as CNN LSTM.

Finally, the recommender component helps users identify suitable stocks based on predictions and their preferences. Backtesting experiments conducted within the study demonstrate 24.5% annualized returns compared to the benchmark (Nifty 50) over the same period of time. Besides, a tool called SHAP can help in analyzing the explainability of predictions through Explainable AI (XAI).

Some limitations associated with the current state of the art can be highlighted. First, the dataset contains only Nifty 50 stocks, and unpredictable events such as geopolitical shocks may lead to inaccurate results.

VIII. Conclusion

To sum up, this paper offers a novel AI/ML framework that utilizes LSTM neural networks, XGBoost algorithms, and multilingual BERT sentiment analysis. With high predictive accuracy and robust features handling capabilities, the framework achieved 95.1% directional prediction accuracy and 1.78 RMSE on January 2015 - November 2025 validation dataset. Additionally, testing the model on early 2026 stock market data, when the Nifty 50 index was closing at 25,713.00 on February 23, 2026, illustrated how effectively the proposed framework copes with forecasting stock prices amid the high volatility affected by global financial trends and market cycles.

As for the recommendation engine, it is capable of generating annualized returns of 24.5%, based on the use of CVaR and graph-based collaborative filtering methods. Therefore, the research clearly proves that combining deep learning, ensemble learning, and sentiment analysis is a good choice for forecasting problems. In comparison to other frameworks, like ARIMA and CNN-LSTM models, the proposed framework allows achieving higher levels of prediction accuracy. Besides, it enables developing more sophisticated models with better financial insights, which also comply with SEBI and RBI guidelines.

Furthermore, apart from technical benefits, the developed framework contributes to expanding the fintech sector in India since it provides a scalable solution for personalized financial advice. With NSE-compliant datasets, the possibility of conducting ablation experiments, and a Flask deployment infrastructure, it allows creating personalized trading platforms. Moreover, this framework supports SHAP-based interpretability for increasing AI model transparency, which is crucial for building trust among users.

Next steps would be to address privacy, scalability, and computation issues associated with using the proposed framework. The integration of federated learning can solve all of these issues as it will be possible to conduct decentralized and privacy-preserving model training in accordance with emerging regulations like Digital Personal Data Protection (DPDP) Act. Moreover, quantum finance will also enable solving more sophisticated problems of portfolio balancing and risk assessment. Finally, multi-asset portfolio optimization is another perspective direction to explore.

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